

LEBRE: Online Forecasting with Evidence-Gated Structural Changes under a Budget of a Few Hundred Operations per Step

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September 2026

Abstract

We describe and evaluate LEBRE, an online one-step-ahead forecaster for a target driven by exogenous inputs, designed to run within a budget of a few hundred floating-point operations per step. The live model is simple: a memory of the target’s own past and an online linear regression on the current inputs, combined by dynamic model averaging. Structural changes run in shadow as experiments: adding, removing or swapping a whole input, or refining its lag response by hierarchical splits of octave bands. A change is accepted only when an anytime-valid e-process of predictive improvement crosses a level set by an online multiple-testing rule. None of these components is new; the contribution is their integration under a fixed compute budget, an explicit condition under which “predictive improvement” may be read as “structure”, and a pre-registered evaluation. We show that when the reference model is itself learned online, its estimation error is predictable from the past. A challenger can therefore improve on it without any new structure, and the structural reading requires small reference misadjustment. In three sequential pre-registered evaluations on never-seen hydrology and building-energy series, the first (125 series) revealed an engineering failure. After the fix, on 40 and 60 series, LEBRE had the lowest error among the low-cost comparators we evaluated (MSE relative to online NLinear 0.812 [95% CI 0.752, 0.866] on 60 series) and no catastrophic series. It tied a SARIMAX with inputs fitted per series and had about 25% higher error than Chronos-2 with covariates. The declared cost contract was narrowly missed (+3.7% mean, +2.0% peak). A C99 port ran on a simulated Cortex-M4F at 3.2–6.0 thousand instructions per step with 73 KB of state. The testing mechanism did not improve accuracy over switching all inputs on; its value is an audited, evidence-backed structure. All code, pre-registrations and results are public.

1 Introduction

Many measurement systems need a forecast of the next value of a target on the device that measures it: the flow at a hydro plant, the load of a building, the output of a feeder. The target is driven by inputs that act with delay (rainfall, temperature, upstream flows). Such a forecaster must be cheap, must work without per-series tuning or offline training, and must not fail badly when the stream has gaps or changes. Foundation models for time series are accurate but cost 10^7 – 10^{10} operations per forecast [2, 11]. Classical models with inputs, such as SARIMAX, need per-series maximum-likelihood fitting [4]. Very small neural forecasters exist [25, 32], but they are usually univariate and batch-trained. Cheap online linear models, finally, do not know by themselves which inputs and delays matter.

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LEBRE takes a specific point in this trade-off. It keeps a simple, cheap live model and treats every *structural* change as an experiment. A challenger that differs from the live model by one change (a whole input, its lag response, or a latent residual state) runs in shadow. The change is accepted only when an anytime-valid e-process [17, 29] comparing the two forecasters [7] crosses a level fixed by an online multiple-testing rule [22, 33]. The result is a forecaster whose structure (which inputs, which delays, since when, with how much evidence) is logged and can be audited.

Contributions. We state them with their limits.

1. **An integrated design** for a forecaster with inputs within a fixed per-step budget. It combines whole-input units with hierarchical lag refinement, learned shadow challengers, tested removal and swap, and online multiple testing (Section 3). The core mechanism is *not* original: accepting structural changes by an e-process of predictive improvement with error control over a stream was proposed for online decision trees by Amoukou et al. [1]. Our contribution is the integration and extension to time series with inputs under a compute budget.
2. **A condition of use** (Section 4). When the reference is learned online, the expected improvement of any predictable challenger is bounded by the reference’s normalised estimation error, and that error is predictable from the past. A test of predictive improvement therefore compares *methods*, not models [15]; it supports a structural reading only when the reference’s misadjustment is small relative to the test margin. We verify this in simulation. The observation follows directly from known results, but we did not find it stated for this setting.
3. **A pre-registered, sequential evaluation** on never-seen series (Sections 5 and 6), including the failure found by the first evaluation and its confirmation on two later ones. It compares LEBRE with budget-class models, classical references and foundation models, and includes a false-change simulation and ablations.
4. **An embedded feasibility check:** a C99 port that reproduces the reference implementation, measured on a simulated Cortex-M4F microcontroller.

We do *not* claim that LEBRE is the most accurate or the most economical forecaster. All statements are relative to the comparators, protocols and data described here.

2 Related work

Lightweight and foundation forecasters. Linear and very small models are strong baselines for long-horizon forecasting: NLinear/DLinear [36], FITS [32] and SparseTSF [25]. They are typically univariate and batch-trained. Pre-trained foundation models range from about 10^6 parameters (TTM, 11) to much larger ones that accept covariates (Chronos-2, 2). On-device incremental learning with exogenous sensors has been demonstrated with small neural networks [24], without statistical control of structure.

Anytime-valid comparison and online multiple testing. E-processes give tests that remain valid under optional stopping [17, 29]. Choe and Ramdas [7] compare two sequential forecasters under a weak null of equal average score, building on the confidence sequences of Howard et al. [20]. LOND-type rules control the false discovery rate online [22], including with e-values under arbitrary dependence [33]. Recent work treats hypotheses whose evidence keeps accumulating [10] and doubly sequential settings [34]. Choi [8] accumulate evidence that a prespecified correction improves a predictive distribution. Jha [23] test Granger non-causality sequentially with family-wise error control.

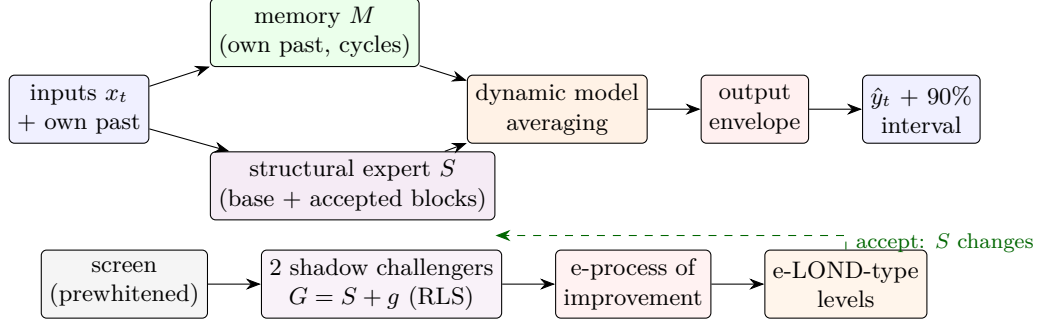


Figure 1: LEBRE. The forecast path (top) is always active and cheap. The change plane (bottom) proposes, tests and decides structural changes; only an accepted change alters S .

Structure selection. Amoukou et al. [1] accept splits of online decision trees by an e-process on the prequential loss difference between incumbent and challenger, with family-wise control over the stream. This is the closest prior work to our mechanism. Recursive sparse identification of ARMAX systems offers asymptotic selection guarantees [14]. Hierarchical testing [26, 35] motivates our parent-before-child refinement of lag bands.

Estimation error in predictive tests. Tests of predictive ability compare forecasting methods, meaning a model together with its estimation procedure [15]. In nested batch comparisons, estimation error penalises the larger model [9]. Han and Qu [18] show that anytime-valid monitors can fire on real streams when the test’s assumptions fail. Fernández-Barrios et al. [13] report incumbent-induced biases in challenger promotion. Our Section 4 makes the estimation-error issue explicit for e-processes that decide structure.

3 Method

Figure 1 summarises the design and Algorithm 1 gives one step. All constants are fixed across series (Table 4); the environment only declares the sampling period(s).

3.1 Live model

Memory M . A regression of the next increment on features of the target’s own past, relative to the last value: reversion to a slow mean, last difference, seasonal naive and one-cycle increment, and a per-phase profile of increments. The features are inspired by the airline model and the Holt–Winters seasonal state [4, 21]. A second declared cycle (the week, for hourly building data) adds two features. Weights are learned by normalised LMS (NLMS) with step 0.05 and scale-relative floors [19, 31].

Structural expert S . A linear regression on the current value of each standardised input and of the target’s own past (the *base*), plus the response blocks of accepted units, learned by NLMS with step $\mu = 0.05$.

Combination. Dynamic model averaging [28]: $w_S = 1/(1 + e^{\eta D_t})$ with $D_t = 0.99 D_{t-1} + [(y_t - \hat{y}_t^S)^2 - (y_t - \hat{y}_t^M)^2]/\hat{\sigma}^2$, $\eta = 0.5$, and $\hat{y}_t = \hat{y}_t^M + w_S(\hat{y}_t^S - \hat{y}_t^M)$. A 90% interval is tracked by coverage feedback in the spirit of Gibbs and Candès [16], without a formal guarantee.

3.2 Units of change and challengers

A *unit* is a whole input (including the target’s own past) or a two-pole latent state of the residual. The response of input i is represented by means of its past over octave bands [1], [2–3], [4–7], [8–15], [16–31]. A band mean dilutes an isolated delay, so the challenger also carries a *peak lag* k^* , which it chooses itself from prewhitened cross-correlations during its warm-up, before any evidence is collected. The challenger for an input fits, by recursive least squares within the episode, the block [band means, $x_{i,t-k^*}, x_{i,t}$] to the residual the change would explain, so the block is judged net of the current value. After acceptance, each band can be *split* into its two halves (a Haar coefficient). A split is a new hypothesis, tested only after its parent was accepted [26, 35]. Change kinds are add, remove (margin $\varepsilon = -0.002$: not worse suffices), swap (when four units are active) and split ($\varepsilon = +0.002$ for add, swap and split).

3.3 Evidence

Let $B_t = 2 \max(\hat{\sigma}_t^S, 0.1 s_t^{\text{slow}})$ be a loss scale floored by the slow scale of the target, and let $\ell_t(u) = \min\{(y_t - u)^2/B_t^2, 1\}$. For a challenger forecast g_t and margin ε , the increments are $d_t = \ell_t(\hat{y}_t^S) - \ell_t(g_t) - \varepsilon$. The weak null of Choe and Ramdas [7] states that the average conditional mean of d_t is ≤ 0 . We use the sub-exponential mixture e-process

$$E_n = \frac{1}{8} \sum_{j=0}^7 \exp \left\{ \lambda_j \sum_{i \leq n} d_i - \psi_{E,c}(\lambda_j) \sum_{i \leq n} (d_i - \gamma_i)^2 \right\}, \quad \psi_{E,c}(\lambda) = \frac{-\log(1 - c\lambda) - c\lambda}{c^2}, \quad (1)$$

with $\lambda_j = 0.9 \cdot 2^{-j}/c$ and predictable centring $\gamma_i = \min(\bar{d}_{i-1}, 0)$.

Lemma 1 (one-sided constant). *Under the weak null with conditional means $\mu_i \leq 0$, if $d_i - \gamma_i \geq -c$ then E_n is upper-bounded by a non-negative supermartingale. Hence $c = 1 + \max(\varepsilon, 0)$ is valid here.*

Proof. The proof in Howard et al. [20, §A.8] uses only the inequality $\exp\{\lambda\xi - \psi_E(\lambda)\xi^2\} \leq 1 + \lambda\xi$ for $\xi \geq -1$, $\lambda \in [0, 1]$ [12]. With $\xi = (d_i - \gamma_i)/c$ and $\lambda \geq 0$, $\mathbb{E}_{i-1}[e^{\lambda(d_i - \gamma_i) - \psi_{E,c}(\lambda)(d_i - \gamma_i)^2}] e^{\lambda(\gamma_i - \mu_i)} \leq (1 + \lambda(\mu_i - \gamma_i))e^{\lambda(\gamma_i - \mu_i)} \leq 1$. Since $\lambda \sum d_i \leq \lambda \sum (d_i - \mu_i)$, the claim follows. Finally, $d_i \geq -(1 + \varepsilon)$ and $\gamma_i \leq 0$ give $d_i - \gamma_i \geq -(1 + \max(\varepsilon, 0))$. $\square$

This is a direct reading of the published proof rather than a new result. The two-sided statement in Choe and Ramdas [7] would use $c = 2(1 + |\varepsilon|)$. In a development simulation (400 trajectories of 5,000 steps, null at the boundary), the one-sided constant kept the type-I error ≤ 0.025 at $\alpha = 0.05$ and raised power from 0.66 to 0.77. Mixtures of e-processes are e-processes, and pauses between episodes form a predictable subsequence, so a hypothesis keeps its evidence across episodes.

3.4 Multiplicity, scheduling and safeguards

The n -th hypothesis receives a level fixed at creation, $\alpha_n = \min(\alpha\gamma_n(R_n + 1), 0.5)$, with $\alpha = 0.05$ and R_n the number of changes accepted so far. Here $\gamma_n = 1/(2P)$ for the first $P = 2 \times \#\text{units}$ hypotheses and a summable tail afterwards. This is the LOND rule with e-values [22, 33]. A change is accepted when $\log E \geq \log(1/\alpha_n)$.

An episode ends on acceptance, when its evidence sum is ≤ 0 after 100 samples (scan rule), or after 5,000 samples. The first 250 samples of an episode only train the challenger. At most two experiments run at a time, and only one warms up at a time. A prewhitened group statistic per inactive input selects what to test (threshold: 99% quantile of χ_5^2); it produces no evidence.

Algorithm 1 One step of LEBRE (simplified; full pseudocode and constants in the research record)

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1:  $\tilde{x}_t \leftarrow$  standardised inputs  $\oplus$  standardised own past; clip to  $\pm 8$  (quarantine)
2:  $\hat{y}^M \leftarrow M(\text{past})$ ;  $\hat{y}^S \leftarrow \beta^\top [1, \tilde{x}_t] + \sum_{u \in \mathcal{A}} w_u^\top \text{block}_u$ ;  $\hat{y} \leftarrow \hat{y}^M + w_S(\hat{y}^S - \hat{y}^M)$ , clipped to the envelope
3: for each running experiment  $e$ :  $g_e \leftarrow \hat{y}^S + \theta_e^\top z_e - (\text{removed contribution})$  ▷ forecast issued here
4: if  $y_t$  observed and not quarantined then
5:   for each experiment  $e$  do
6:     if past warm-up then observe  $d = \ell(\hat{y}^S) - \ell(g_e) - \varepsilon_e$  in its e-process
7:     update  $\theta_e$  by RLS; during warm-up accumulate the peak-lag statistic
8:     NLMS on  $\beta$ ,  $\{w_u\}$  with residual  $y_t - \hat{y}^S$ ; update the screening statistics
9: update  $D$ ,  $w_S$ , interval; memory gets  $y_t$  (or the last observed value during a gap)
10: if  $t \bmod 10 = 0$  then accept / close experiments; open new ones (removal tests first, then best screened unit or split)

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Two output safeguards act only on the forecast path. During target gaps the memory receives the last observed value, never its own forecast. The forecast is also clipped to an envelope of the observed values widened by half its width. Both were added after the first evaluation (Section 6.1).

3.5 Cost

Operations are counted in the code, component by component. The per-step cost is almost fixed: memory, base and screening always run, experiments add cost only when a slot is busy, and peaks occur at decision steps. The contract declared during development was mean ≤ 400 and peak $\leq 1,000$ operations per step on real series. It was revised once, from 250, before the confirmatory evaluations.

4 A condition of use: misadjustment of the reference

Let $\mathcal{F}_{t-1}$ be the information before y_t , $f_t^* = \mathbb{E}[y_t \mid \mathcal{F}_{t-1}]$, $\epsilon_t = y_t - f_t^*$, and let the reference S_t , the challenger $G_t = S_t + g_t$ and the scale B_t be predictable. Define the reference's estimation error $\eta_t = f_t^* - S_t$, which is predictable, and $\Delta_t = [(y_t - S_t)^2 - (y_t - G_t)^2] / B_t^2$.

Proposition 1. $\mathbb{E}[\Delta_t \mid \mathcal{F}_{t-1}] = (2g_t\eta_t - g_t^2) / B_t^2 \leq \eta_t^2 / B_t^2$, with equality at $g_t = \eta_t$.

Proof. $(\epsilon_t + \eta_t)^2 - (\epsilon_t + \eta_t - g_t)^2 = 2g_t(\epsilon_t + \eta_t) - g_t^2$ and $\mathbb{E}[\epsilon_t \mid \mathcal{F}_{t-1}] = 0$; maximise over g_t . □

Corollary 1. *If $\frac{1}{n} \sum_{i \leq n} \eta_i^2 / B_i^2 \leq \varepsilon$ for all n , then under the structural null (the unit's features carry no information about y), no predictable challenger has positive mean improvement beyond the margin. The test's null then holds, and its error control transfers to the structural null.*

For NLMS with misadjustment $M \approx \mu / (2 - \mu)$ and $B_t \approx 2\hat{\sigma}$, the corollary requires $M \lesssim 4\varepsilon$, i.e. $\mu \lesssim 0.016$ at $\varepsilon = 0.002$.

A challenger with a coefficient that is constant over a window can capture at most the part D_u of η^2 / B^2 projected on its features. As an approximation, the log-evidence grows like $nD_u^2 / (2\sigma_d^2)$. False acceptances are therefore unlikely within a series of length T when $D_u < \sigma_d \sqrt{2 \log(1/\alpha_k)} / T$ (*practical regime*). This is not a guarantee.

Verification. With the reference learned by NLMS at $\mu = 0.1$, 10 of 40 structureless targets had a change accepted. It was always the unit with the largest projection of η , the target's own past. At $\mu = 0.05$ and $\mu = 0.03$ there were 0 and 0 of 40 (Figure 2). LEBRE uses $\mu = 0.05$, which is in

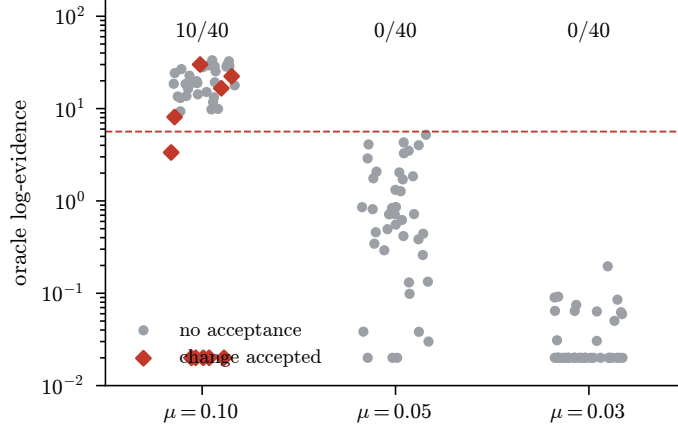


Figure 2: Structureless targets with real inputs (development, 120 runs). Log-evidence of the best fixed-coefficient challenger (projection oracle) versus the acceptance threshold (dashed); red diamonds: runs where the model actually accepted a change (for those, the oracle was measured after the acceptance and can sit at the floor); numbers: accepted runs. The median oracle crosses the threshold at $\mu = 0.1$ (18.2) and stays below it at $\mu = 0.05$ (maximum 5.2).

the practical regime, *not* in the exact one. False-change control is therefore checked empirically (Section 6.4).

5 Experimental protocol

Task and metric. Online one-step-ahead forecasting with the current inputs available; all models receive the same causally standardised inputs. The metric is the mean squared error on the last 70% of each series (observed target), divided by that of an online NLinear calibrated on the same series, with the geometric mean across series. 95% CIs come from a bootstrap stratified by domain (2,000 resamples, fixed seed). A series is *catastrophic* when the relative error exceeds 10, the forecast deviates by more than 10 times the observed range, or the forecast is non-finite.

Data. Hourly and daily hydro data from the Brazilian grid operator (ONS; cascaded plants, inputs = upstream flows), catchments of CAMELS-BR [6] (inputs: precipitation, evapotranspiration, temperature), and heating and cooling meters of the Building Data Genome 2 [27] (inputs: weather). Series were split by seeded permutations, before any modelling, into development and reserves, and the loader refuses reserve series outside a pre-registered run. Development also used two system-identification benchmarks and declared synthetic suites.

Pre-registration. Before each reserve was accessed, a document fixed the questions, metrics, comparators, criteria and code hashes. Each reserve was run once. Reserve 1 had 125 series (ONS 45, CAMELS-BR 50, BDG2 30), reserve 2 had 40 (CAMELS-BR 20, BDG2 20) and reserve 3 had 60 (CAMELS-BR 30, BDG2 30); the ONS rivers were exhausted after reserve 1.

Comparators. The *budget class* (up to $\sim 10^4$ operations per forecast) contains:

- online NLinear and DLinear [36];
- online Holt–Winters;
- a dense ARX learned by NLMS and an online LASSO;

Table 1: Reserve 3, full test region (60 series). Relative MSE (geometric mean; lower is better), share of series with relative error above 1.5, catastrophic series and analytic operations per forecast.

Model	Class	CAMELS	BDG2	All [95% CI]	> 1.5	Cat.	Ops/forecast
LEBRE	budget	0.858	0.768	0.812 [0.752, 0.866]	0%	0	415
SARIMAX with inputs	reference	0.817	0.803	0.810 [0.763, 0.851]	0%	0	ML fit
Online NLinear	budget	1.000	1.000	1.000 [1.000, 1.000]	0%	0	674
Online DLinear	budget	1.076	0.947	1.009 [0.948, 1.066]	3%	0	1,533
Online Holt-Winters	budget	1.213	1.244	1.229 [1.136, 1.323]	25%	0	20
FITS	budget	1.362	1.137	1.244 [1.158, 1.335]	18%	0	9,757
Least-squares ARX	reference	0.677	2.867	1.394 [0.998, 2.170]	20%	4	48,944
Dense ARX (NLMS)	budget	1.008	2.711	1.653 [1.184, 2.730]	38%	2	474
Online LASSO	budget	1.015	3.048	1.759 [1.237, 2.898]	38%	4	705
LEBRE v0.51	budget	2.180	2.140	2.160 [1.440, 3.723]	48%	4	108
SparseTSF	budget	2.980	2.886	2.933 [2.512, 3.446]	87%	3	4,016

- FITS [32] and SparseTSF [25], fitted on the first 15% and then frozen;
- the previous LEBRE version (v0.51) as an ablation.

References (ceilings, not opponents) are a SARIMAX with inputs fitted per series by maximum likelihood, a least-squares ARX with online order selection, TTM zero-shot and fine-tuned with the inputs as exogenous channels [11], and Chronos-2 with covariates [2]. Foundation models are too expensive to run at every step. They were therefore evaluated on 1,000 points per series, the same points for all models, with the current inputs given as known future covariates.

Microcontroller. The canonical configuration was ported to C99 (float64 and float32) and compiled for a Cortex-M4F with `arm-none-eabi-gcc` 15.2 (-O2, single-precision FPU, no FMA contraction). It was run in Renode 1.17 [3] on an STM32F4 model, with the DWT cycle counter configured to count executed instructions. Cycles were estimated from execution traces with the published Cortex-M4 timings (zero-wait flash assumed). No physical board was used and energy was not measured.

6 Results

6.1 Reserve 1: an engineering failure

The version evaluated in reserve 1 had no output safeguards, and 6 of 125 series exploded (relative error up to 10^{79}). During long target gaps the memory filled its own history with its own forecasts, a recursion that some learned weights made unstable. The pre-registered result was dominated by these series (geometric mean 5.20). In an exploratory, non-pre-registered view without them, the relative error was 0.706. The safeguards of Section 3 were designed after seeing this failure, verified by a development stress test, and then evaluated on two new reserves. The fix was therefore not designed blind.

6.2 Reserves 2 and 3

Tables 1 and 2 and Figure 3 report the pre-registered results of reserve 3; reserve 2 was consistent. The findings are:

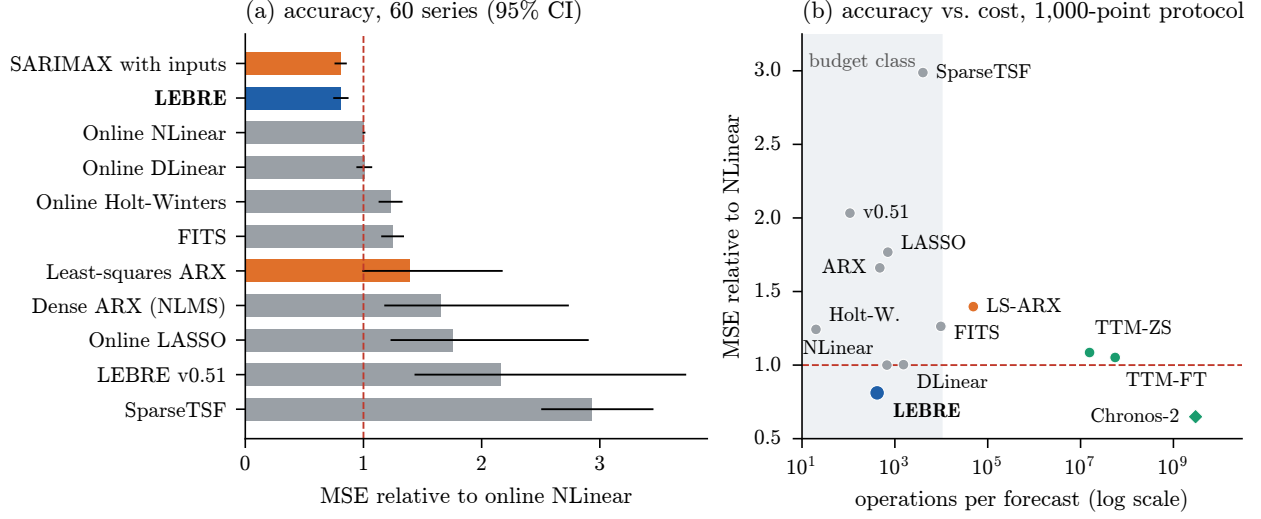


Figure 3: Reserve 3 (60 never-seen series). (a) MSE relative to online NLinear on the full test region, geometric mean with 95% CIs (blue: LEBRE; grey: budget class; orange: classical references). (b) Error on the 1,000-point protocol versus operations per forecast (green: foundation models); the shaded band is the budget class. SARIMAX is omitted on the right because its per-forecast cost was not counted. The Chronos-2 cost (diamond) is an order-of-magnitude estimate.

Table 2: Reserve 3, 1,000-point protocol with foundation models. Paired ratio LEBRE/model (geometric mean over series, 95% CI); below 1 means LEBRE errs less.

Model	CAMELS	BDG2	All	Paired ratio	LEBRE wins	Ops/forecast
LEBRE	0.836	0.787	0.811	–	–	≈415
SARIMAX with inputs	0.798	0.818	0.808	1.004 [0.956, 1.056]	28/60	ML fit
TTM zero-shot	1.204	0.978	1.085	0.747 [0.704, 0.795]	57/60	15.6M
TTM fine-tuned (inputs)	1.049	1.054	1.052	0.771 [0.699, 0.844]	50/60	55.6M
Chronos-2 (covariates)	0.782	0.539	0.649	1.250 [1.178, 1.337]	9/60	~1e9–1e10

- **Budget class.** LEBRE had the lowest error among the budget-class comparators: 0.812 [0.752, 0.866] on reserve 3 and 0.767 [0.657, 0.858] on reserve 2. It had no catastrophic series and none above 1.5 in either reserve. Paired ratios against online DLinear and FITS were 0.804 and 0.652.
- **Ultralight models in this regime.** FITS and SparseTSF, used here outside their design regime (one-step, frozen after fitting on 15%), were behind online NLinear.
- **Classical reference.** LEBRE tied a SARIMAX with inputs fitted per series. The paired ratio was 1.002 [0.966, 1.041] on reserve 3 and 0.999 on reserve 2.
- **Foundation models.** LEBRE’s error was lower than TTM’s, with ratios 0.747 (zero-shot) and 0.771 (fine-tuned); these depend on the fine-tuning setup we chose in development. It was higher than that of Chronos-2 with covariates, with ratios 1.250 [1.178, 1.337] on reserve 3 and 1.161 on reserve 2. The largest gap was on buildings (0.787 vs. 0.539).
- **Interval.** The 90% interval had empirical coverage 0.904.
- **Cost.** The analytic cost averaged 415 operations per step, with a maximum of 1,020. The contract declared beforehand was therefore **not met**, exceeded by 3.7% in the mean and 2.0% in the peak.

Table 3: Simulated Cortex-M4F (float32 port): instructions per step, estimated time at 168 MHz (CPI 1.6), test-region NMSE (identical to Python) and accepted changes. Rows 4–7 belong to the pre-registered reserve 3.

Series	Inputs	Mean	Median	p99.9	Peak	μ s	NMSE	Changes
ONS (dev.)	1	3,703	3,363	12,327	42,021	35	0.2826	4
CAMELS (dev.)	3	4,560	4,599	14,387	48,447	43	0.0229	3
BDG2 (dev.)	2	5,964	5,607	16,843	45,654	57	0.0305	2
CAMELS 51795000	3	3,203	3,003	12,223	36,771	31	0.1465	0
CAMELS 64625000	3	3,521	3,215	12,267	36,708	34	0.3249	0
BDG2 Fox_education_Ollie	2	5,806	5,587	15,375	45,402	55	0.0162	3
BDG2 Panther_education_Vincent	2	5,528	5,463	16,971	45,696	53	0.0584	3

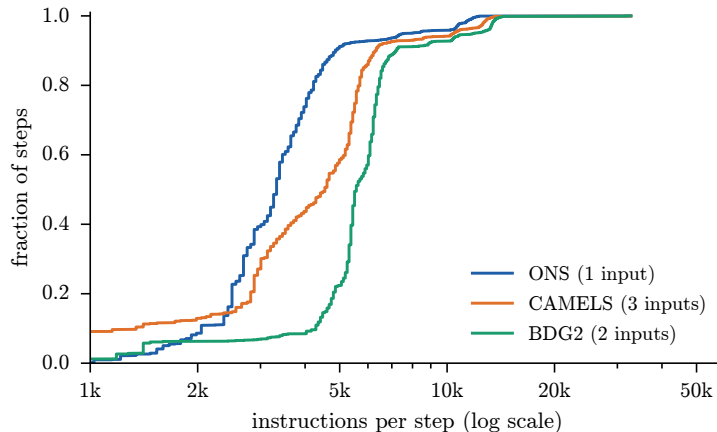


Figure 4: Cumulative distribution of instructions per step on three development series (simulated Cortex-M4F). The cheapest steps have a missing target.

6.3 C99 port and simulated microcontroller

The float64 C build reproduced the Python decision sequences on 29/29 development series. The float32 build reproduced them on 28/29 development and 54/60 reserve-3 series; in the others a decision near the threshold shifted by a few steps. On the simulated Cortex-M4F, the mean cost was 3.2–6.0 thousand instructions per step (Table 3 and Figure 4). Peaks of 37–48 thousand instructions occurred only at decision steps. The estimated CPI was 1.62 and 1.57, giving about 31–57 μ s per step at 168 MHz. The model state takes 73 KB, dominated by a fixed-capacity hypothesis table: this fits an STM32F407 but not smaller microcontrollers without reducing capacities.

6.4 False-change control under dependence (simulation)

We ran the frozen model on 700 synthetic runs of 30,000 steps (plan and criterion declared beforehand). The runs used four correlated inputs with persistence 0.5 or 0.95, four null scenarios and three scenarios with one true input. The nulls were white noise, an autoregressive target, noise whose variance depends on an input, and a variance break. The scenarios with a true input were plain, with a decoy correlated 0.7 with the true input, and with heavy-tailed autoregressive noise.

- **Null scenarios.** 0 of 400 runs accepted any change.
- **Scenarios with a true input.** By the declared per-scenario criterion, all stayed within $\alpha = 0.05$. With persistence 0.5, however, two cells exceeded it: FDR 0.090 (decoy, 7/50 runs) and 0.077

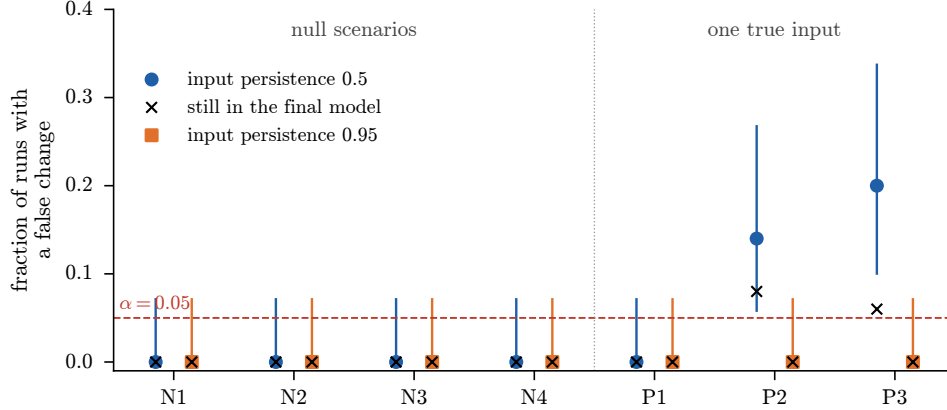


Figure 5: False-change simulation (700 runs; 50 per scenario and input persistence). Markers: fraction of runs with at least one false change, with Clopper–Pearson 95% intervals; crosses: fraction where the false change was still in the final model.

(dependent noise, 10/50) (Figure 5).

In all 17 affected runs, the false change was an input *correlated* with the true one and accepted before it. While the true input is absent, such a substitute does improve the reference forecast: the test answers its own question correctly, but the answer is not structural. The periodic removal test took the substitute out in 10 of 17 cases; in 7 of 700 runs it stayed in the final model. An accepted input therefore means “improved the forecast of the current model, with evidence”, not “is the cause”.

6.5 Ablations

- **Testing mechanism.** Switching all units on from the start, without experiments, gave the same accuracy. On reserves 2 and 3 together (100 series, exploratory, because the reserves had already been used) the paired ratio LEBRE/all-on was 0.988 [0.970, 1.006], and the all-on version costs less (395 vs. 415 operations per step). The testing mechanism is therefore not justified by accuracy: its role is an evidence-backed, logged structure, and its practical value for a downstream decision was *not* demonstrated.
- **Output safeguards.** Without them, reserve 2 had 2 catastrophic series and a relative error of 1.107 (paired ratio 0.693 in favour of the safeguards).
- **Other components.** Hierarchical splits made no difference on real data (ratio 0.996). Per-lag hypotheses (about 400 per series) instead of whole-input units were worse in development (0.757 vs. 0.645 on 1,000 points).

7 Limitations

- **Scope.** Two domains, 1–5 inputs, one step ahead with current inputs available; the last two reserves have no ONS series. With 40 and 60 series the confidence intervals are wide.
- **Statistics.** The e-values are valid under the stated assumptions, but with the adaptive reference false-change control is only empirical (Sections 4 and 6.4). Correlated substitutes can be accepted, near-identical inputs are indistinguishable, and very smooth inputs with medium or long delays are often not discovered. The last point is consistent with known power losses for persistent regressors [5, 30]. Removal tests rarely accept: with margin -0.002 the evidence for removing a

useless unit grows slowly relative to the episode length.

- **Comparisons.** SARIMAX, FITS, SparseTSF and TTM were fitted once in batch; online ARMAX/Kalman variants were not evaluated. The fine-tuning of TTM used one configuration chosen in development. Foundation models were evaluated on 1,000 points per series.
- **Cost.** The declared contract was not met. The microcontroller results come from a simulator, cycles are estimated from traces, and energy was not measured.
- **Process.** The output safeguards were designed after the failure of reserve 1. The cost contract was revised once during development. The literature search behind Section 2 used one search engine and mostly abstracts.

8 Discussion and conclusion

Within the budget of a few hundred operations per step and in the tested domains, a simple online linear model with a seasonal memory and output safeguards delivered competitive one-step forecasts without catastrophic failures. It tied a per-series SARIMAX and stayed within about 25% of a large foundation model with covariates. The evidence-gated structural mechanism did not add accuracy. What it adds is an auditable record of which inputs and delays entered the model, when and with what evidence. We also make explicit a condition under which such a record may be read structurally: the reference that challengers are compared against must have small misadjustment. When correlated substitutes exist, even that reading is only about predictive relevance. Natural next steps are:

- a reference with provably small misadjustment (a slow shadow reference);
- online ARMAX/Kalman comparators;
- a physical board with energy measurement;
- a smaller state;
- evaluation in further domains;
- a demonstration that the logged structure improves a downstream decision.

Reproducibility statement

The complete research record is public at <https://github.com/Iquitim/lebre-research> and archived as [doi:10.5281/zenodo.23049103](https://doi.org/10.5281/zenodo.23049103). It contains the Python reference implementation, the C99 port and firmware, all experiment scripts, the pre-registrations with their hashes, per-series result tables, reports and SHA-256 manifests of every frozen version. Every number in this paper is generated from result files by a script (`paper/build_assets.py`). A smoke test re-runs the frozen model on reserve series and reproduces the published predictions bit for bit. Raw third-party data and per-series prediction arrays are listed with checksums, not redistributed. An installable library implementing exactly this version, and tested bit-for-bit against it, is in `packages/lebre`.

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A Constants

Table 4: Constants of the evaluated configuration (identical for all series).

Group	Constants
Units	one per input (incl. the target’s own past) + residual latent state (poles 0.8, 0.95); at most 4 active
Response	bands [1], [2–3], [4–7], [8–15], [16–31] + peak lag (chosen after 200 samples) + current value; splits into halves
Challengers	RLS within episode, $P_0 = 10I$; warm-up 250; 2 slots; one warm-up at a time
Evidence	$\ell = \min\{e^2/B^2, 1\}$, $B = 2 \max(\hat{\sigma}_S, 0.1s^{\text{slow}})$; $c = 1 + \max(\varepsilon, 0)$; 8 values $\lambda_j = 0.9 \cdot 2^{-j}/c$; $\varepsilon = +0.002$ (add, swap, split), -0.002 (remove)
Multiplicity	$\alpha = 0.05$; $\gamma = 1/(2P)$ for the first $P = 2 \cdot \#\text{units}$, then $0.5 \gamma_k \propto 1/(k \log^2 k)$; level $\min(\alpha\gamma(R+1), 0.5)$
Episodes	scan rule after 100 samples; cap 5,000; cool-down 2,000; removal test every 2,000; decisions every 10 steps
Screening	prewhitened AR(1), EMA weight 0.02, every 4 steps per part; thresholds 15.09 (χ_5^2), 6.63 (χ_1^2)
Base of S	NLMS $\mu = 0.05$; input contract $\pm 8\sigma$ with 31-step quarantine
Memory	NLMS $\mu = 0.05$; slow mean 0.01; profile 0.1; floors 10^{-3} , 10^{-4} ; declared cycle(s); hold during gaps
Combination, output	$\lambda = 0.99$, $\eta = 0.5$, every 8 steps; envelope rate 10^{-3} , margin $0.5\times$
Interval	target 90%, gain 0.1, every 8 steps
Inputs	causal standardisation, exponential rate 10^{-4} from mean 0 / variance 1